

Using ResearchGPT for learning Active Inference and doing Ant research

9:25

all right the goal of this video is to go through a conversation that I just had with research GPT by siace at typeset. to demonstrate some capacities of this model and also learn about active inference ants and Beyond the prompts that I wrote will be read and answers will only be overviewed pause the video or see the sharable conversation Link in the video description for Fuller text so I jumped right in with the prompt provide a comprehensive account of the relationship between active inference and the free energy principle it provided five relevant citations and so I wondered about this in the context of developing a PhD dissertation around the philosophical articulations of active inference and the free energy principle to go beyond just literature search and see if it could output a full dissertation it searched and again returned a list of five papers which was not exactly what I was looking for but they are relevant papers I then pulled into a more beginner or learner line of questioning can you provide a bullet point list of the 31 most important terms for my total comprehensive understanding of active inference it could be technical metaphysical pragmatic or whatever kind of terminology is going to catalyze this Today's starting point of my curiosity and the terms that it recommended were

active inference free energy principle
basy and brain predictive coding marov
blanket variational free energy
generative model surprise Precision
belief updating epistemic action entropy
predictive error perceptual inference
hierarchical models action selection
propri acception affordances embodied
cognition neurophenomenology decision
Theory optimal control theory
homeostasis allostasis interception
empirical base metacognition cognitive
dissonance in activism sensory
attenuation phenomenal

Consciousness I wrote seeing all these
terms makes me truly excited however I
am not sure where or how to begin based
upon evidence supported techniques and
learn learning theory can you write me a
pseudo code loop that I can follow
during the day while I study active
inference it provided the pseudo code
the study session is initialized and
three study sessions occur per day
during each study session start with A
Brief Review focus on one key term or
concept read and understand the concept
apply Active Learning techniques
reflecting and mind maps relate the New
Concept to an already learned concept
test understanding take a short break
repeat with a new concept or deepen
understanding of the same concept and
the session reflect and plan take a
longer break to help with memory
consolidation increment the study
session and reflect and plan at the end
of the day optional to engage in related

discussions or online forums for

Community

Learning and that is

explained excellent already I feel my

confidence increasing and veilance

rising and I did can you provide

citations that address specifically

professionally how active inference and

reinforcement learning are similar and

different several extremely relevant

papers including probably the most

relevant paper are provided so five

relevant papers great work those are

some of the right citations for us to

consider can you summarize similarities

and differences between active inference

and reinforcement learning I am looking

for a bullet point list that contains as

many lines as needed such that every

line is composed to be semantically

univocal and among lines the entire

semantic space of the best possible

answer is surveyed thank you and a list

was provided with similarities seven of

them and 11 differences between active

inference and reinforcement learning one

could probably quibble or update or lump

or split but overall it did really

well so I wanted to turn to the ant

colony setting given all the above and

alluding to contemporary discourse on

applied Act of inference can you write

me 23 specific situations where testable

hypotheses arise from bare observation

of ant colony behavior and 23 situations

were provided different situations that

ant colonies engage in trail formation

division of labor brood sorting so

on I wrote Let's delve into idea 7 nest building this concept is very interesting to me and a few of my nestmates as well can you do literature meta analysis into ant nest building and report back with an elegant blend of the most cited papers and the most relevant ones papers were returned 10 papers were returned I don't know which ones are highly cited and which ones are relevant there were some cut off answers but broadly speaking this looks pretty relevant since it's dealing with empirical and review work on ant nests so I wrote great can you write out the comprehensive pseudo code for starting out an agent-based active inference model which captures most of the relevant terminology from all the ant papers above and it did this the ant Behavior studies parameters were initialized the ant agent was defined in terms of its perception of the environment and action selection sense making and decision- making that is based upon minimization of free energy of course it's not fully written out here in the pseudo code but that's something that one could go further into a discussion with model or use standardized methods action performance and communication initialization running the simulation updating the environment optional introduction of environmental changes and maybe a finalized simulation routine that summarizes or gives

visuals excellent I'm excited about this
pseudo code which I summarized as an
active inference model of within within
and among task switching extra word
there as being like the semantic kernel
or strange attractor of my
PhD I'm going to provide you my first
ideas for this seven chapter
dissertation I'm thinking of after
describing this sketch to you and
absolutely including all of the above
discussion we've had please
comprehensively expand this into a table
where the columns are chapter number
chapter name chapter goal and chapter
methodology
first chapter more about literature
review use social task allocation and
active inference second chapter
proposing mathematical framework for
infra ANS task allocation third chapter
developing a more physiological grounded
extension for future integration with
multiomic data fourth chapter is a
collaboration with friend Who develops
secure autonomous video deployments
where we create a cheap and eco-friendly
neuromorphic ant colony Behavior
recording device that fits inside of a
small backpack fifth chapter explicitly
tackles the metaphysical and onto
linguistic dimensions of the project
sixth chapter describes the synthetic
intelligence system enabling all the
chapters in terms of requirements
engineering seventh and final chapter
encapsulates the entire dissertation
with several important discussions

connections and insights otherwise inappropriate for the six previous chapters and it provides those chapters six of them in a table I wrote wonderful please improve this table so that the existing insights are crystallized any missing relevant features are added and also add a new column for chapter relevance as distinctly articulated from chapter goal the chapter relevance column was provided and it does seem to differ from the goal and now just to close out this conversation and video I wrote last question for now can you come up with 27 hilarious possible titles for the overall project ideally including some acronym which spells out an ant related pun from the full technical title go wild thank you and I'll just read out the puns Antics btics antidote antsy antenna neuro Antics I ant intelligent Ant Farm analytic Quant an sin antic infra ant anticipate mbul an ant Venture cogn ant wander ant Dina ant Converse an frantic benevolent elegant Fon an consultant omnis an demsin ant ramp ant these titles blend humor with technical aspects of your project incorporating ant related puns and acronyms that reflect the essence of your research in a playful manner so this is just a first pass on the research GPT tool looks like great work by the SI base team and just very interesting and exciting as we all head into this

scientific environment together all
right see you around the Institute
bye